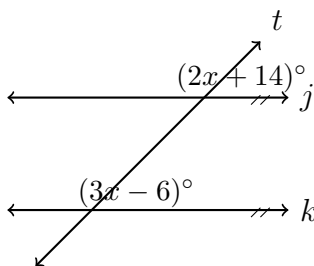


Solving for Angle Measures on Parallel Lines

congruent angle pairs, supplementary angle pairs, and two-step chains

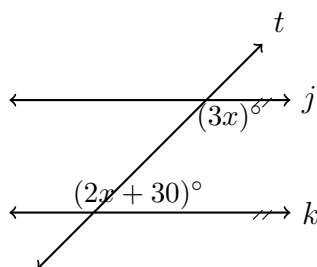
- Double arrowheads mark the parallel lines. Name the angle pair first, then write the equation the name gives you: congruent pairs are *equal*; supplementary pairs *add to 180*.
 - Congruent on parallel lines: corresponding, alternate interior, alternate exterior. Supplementary: same-side interior, and any linear pair.
 - Show the equation before you solve it, and check your value by substituting back.
-

1. $j \parallel k$. The two marked angles are corresponding angles. Find x .



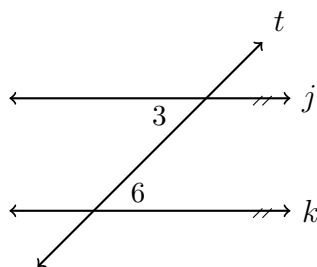
Answer: _____

2. $j \parallel k$. The two marked angles are same-side interior angles. Find x .



Answer: _____

3. $j \parallel k$, and $\angle 3$ and $\angle 6$ are alternate interior angles with $m\angle 3 = (5x - 12)^\circ$ and $m\angle 6 = (3x + 18)^\circ$.



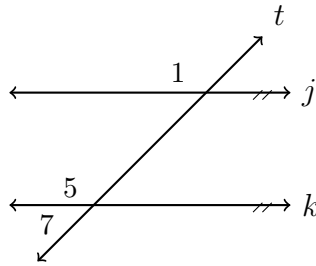
- (a) Find x .

Answer: _____

- (b) Find $m\angle 3$.

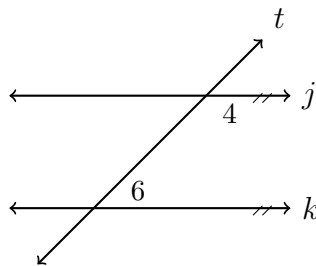
Answer: _____ degrees

4. $j \parallel k$ and $m\angle 1 = 47$ degrees. Find $m\angle 7$. (Move first to the angle corresponding to $\angle 1$, then around that vertex.)



Answer: _____ degrees

5. $j \parallel k$, and $\angle 4$ and $\angle 6$ are same-side interior angles with $m\angle 4 = (4x - 5)^\circ$ and $m\angle 6 = (6x + 15)^\circ$.



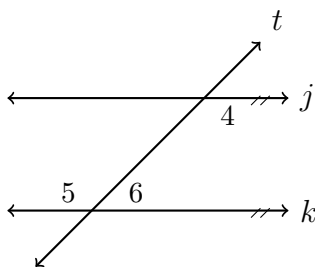
- (a) Find x .

Answer: _____

- (b) Find $m\angle 4$.

Answer: _____ degrees

6. $j \parallel k$, with $m\angle 4 = (3x + 5)^\circ$ and $m\angle 5 = (x + 45)^\circ$.



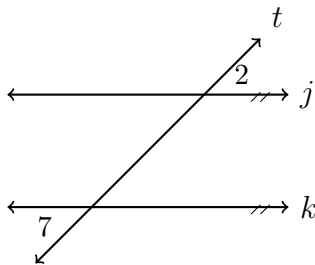
- (a) $\angle 4$ and $\angle 5$ are alternate interior angles. Find x .

Answer: _____

- (b) Find $m\angle 6$.

Answer: _____ degrees

7. $j \parallel k$, and $\angle 2$ and $\angle 7$ are alternate exterior angles with $m\angle 7 = (8x - 20)^\circ$ and $m\angle 2 = (5x + 25)^\circ$.



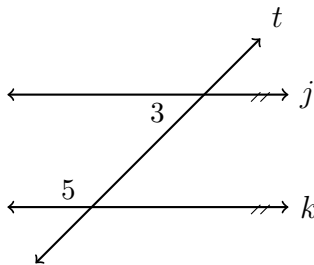
- (a) Find x .

Answer: _____

- (b) Find $m\angle 2$.

Answer: _____ degrees

8. $j \parallel k$, and $\angle 3$ and $\angle 5$ are same-side interior angles with $m\angle 3 = (2x + 8)^\circ$ and $m\angle 5 = (4x + 4)^\circ$.



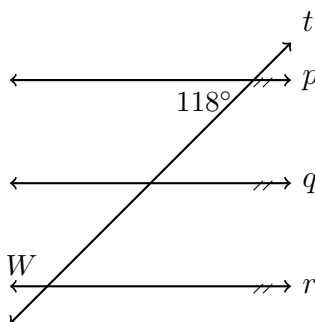
- (a) Find x .

Answer: _____

- (b) Find $m\angle 5$.

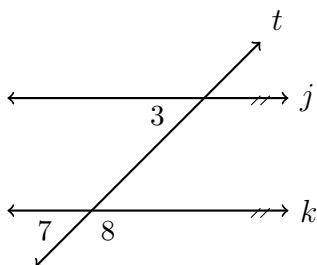
Answer: _____ degrees

9. Three parallel lines p , q and r are cut by one transversal t . The marked angle at p measures 118 degrees. Find $m\angle W$, the marked angle at r .



Answer: _____ degrees

10. $j \parallel k$, with $m\angle 3 = (5x + 3)^\circ$ and $m\angle 8 = (7x - 27)^\circ$.



- (a) Name the two relationships you chain to get from $\angle 3$ to $\angle 8$, and explain why they make $\angle 3$ and $\angle 8$ supplementary rather than congruent.

- (b) Find x .

Answer: _____

- (c) Find $m\angle 8$.

Answer: _____ degrees